



Synthesis of $\text{Ca}_2\text{SiO}_4:\text{Dy}^{3+}$ phosphors from agricultural waste for solid state lighting applications



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ABSTRACT

Trivalent dysprosium (Dy^{3+}) ions -doped calcium silicate (Ca_2SiO_4) phosphors have been synthesized by utilizing agricultural waste of egg shell and rice husk through solid-state reaction method. The synthesized Ca_2SiO_4 powders thus obtained are crystallized in monoclinic structure with unit cell parameters of $a = 5.53 \text{ \AA}$; $b = 6.67 \text{ \AA}$; $c = 9.13 \text{ \AA}$; $\beta = 87.43^\circ$ and irregular shape morphology. Luminescent properties of $\text{Dy}^{3+}:\text{Ca}_2\text{SiO}_4$ phosphors were studied by varying active ion concentration. The phosphors emit characteristic blue and yellow emissions of Dy^{3+} ions corresponding to the $^4\text{F}_{9/2} \rightarrow ^6\text{H}_{15/2}$ and $^4\text{F}_{9/2} \rightarrow ^6\text{H}_{13/2}$ transitions, respectively. Color coordinates evaluated from emission spectra are found to fall in the white light region. Decay curves for the $^4\text{F}_{9/2}$ level of Dy^{3+} ions exhibit single exponential nature and turn into non-exponential with shortening of lifetime from 739 μs to 510 μs when Dy^{3+} ion concentration is increased from 0.001 to 0.5 mol%. All these results confirm that $\text{Ca}_2\text{SiO}_4:\text{Dy}^{3+}$ phosphors are suitable for the use as low cost white light emitting phosphors.

1. Introduction

In recent years, the conversion of agricultural waste into low cost luminescent materials is a challenging task to reduce environmental pollution by converting waste into useful low cost materials [1–8]. The trivalent lanthanide (Ln^{3+}) ions-doped luminescent materials have attracted much attention due to their superb applications in the fields of lighting, display, optical sensors, solar energy harvesting, bio-imaging, and so on [9–14], in particular for the development of light emitting devices. Among all the light emitting devices (LEDs), the white light emitting devices (WLEDs) are deemed to become important sources because of their low power consumption, higher brightness, longer durability, fast response, environment benign, etc. [15–17]. The WLEDs are normally fabricated by the combination of a blue LED with a yellow emitting phosphor ($\text{YAG}:\text{Ce}^{3+}$). But the yellow phosphor show chemical instability, low absorption cross-section and moreover they are synthesized with commercial chemicals [18,19]. Hence, the quest to identify/develop new phosphors with low cost, chemically stable and high luminescence efficiency is a daunting task.

Hence, the present investigation is aimed at synthesis of low cost white light emitting phosphors by utilizing agricultural waste materials such as rice husk and egg shells. Nowadays recycling of agricultural residues such as rice husk and egg shells into luminescent materials is

attracting much attention as there is twofold advantage in it. One is prevention of environmental pollution and second one is development of luminescent materials at the low cost. Therefore, the main objective of the present study is not only to reduce environmental pollution but also to make use of them effectively for the development of phosphors with better luminescence properties by using agricultural waste materials (rice husk and egg shells). Rice husk is a renewable agricultural material and contains an active form of silica (SiO_2) of about 98%, substitute for commercial SiO_2 . Another key waste material is egg shell (ES), which is comprised of calcium oxide (CaO) of about 96%, substitute for commercial CaO.

Among Ln^{3+} ions, Dy^{3+} ion is capable of emitting a white light since its emission spectrum consists of blue and yellow emissions [20–36]. The $^4\text{F}_{9/2} \rightarrow ^6\text{H}_{13/2}$ transition is hypersensitive in nature and its emission intensity strongly depends on the local environment whereas the $^4\text{F}_{9/2} \rightarrow ^6\text{H}_{15/2}$ transition is magnetic-dipole in nature and its intensity is almost independent of surrounding environment. Hence, Dy^{3+} -doped phosphors are capable of emitting white light by varying the chemical composition of the host.

In recent years, RE^{3+} -doped phosphors such as vanadates [20], molybdates [37], tungstates [38,39], silicates [40], and aluminates [41,42] besides RE^{3+} -doped glasses [21,35,43] have been investigated. Among these host materials, silicates are attractive class of inorganic

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phosphors due to their higher physical and chemical stabilities, and wide transparency [44]. The main objective of the present investigation is synthesis of Dy^{3+} -doped Ca_2SiO_4 phosphors using the agricultural waste materials such as rice husk ash and egg shells and study their structural, morphological and luminescent properties for the development of low cost white light emitting phosphors.

2. Experimental procedure

$\text{Ca}_2\text{SiO}_4:\text{x}\text{Dy}^{3+}$ ($\text{x} = 0.001, 0.005, 0.01, 0.05, 0.1, 0.2, 0.3, 0.4$ and $0.5 \text{ mol}\%$) phosphors were prepared by a conventional solid-state reaction method in three steps and the detailed procedure is shown in the following sections.

2.1. Synthesis of SiO_2 from rice husk

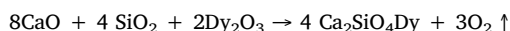
Rice husk (RH) was washed with normal tap water for removing soil and dust. After that RH was dried at normal room temperature for 24 h, and placed uniformly in aluminum rectangular tray with large surface area. Flat trays were chosen for uniform distribution of rice husk powder so as to have sufficient air circulation. The tray containing RH was placed in a furnace at 600°C for 6 h and then 1000°C for 8 h for sintering, finally white rice husk ash (RHA) was obtained which was grinded very well to obtain fine powder.

2.2. Synthesis of CaO from egg shells

White egg shells (ES) were washed with tap water to remove all adhering albumen, dried at room temperature for 12 h, after that the ES were crushed into small pieces. These crushed ES were placed uniformly in aluminum rectangular tray with large surface area for sufficient air circulation. The tray containing ES was placed in a furnace at 900°C for 4 h for sintering and finally white ES powder was obtained and then well grinded to obtain fine powder.

2.3. Synthesis of Dy^{3+} -doped Ca_2SiO_4 phosphors

Calcium silicate (Ca_2SiO_4) phosphors doped with Dy_2O_3 ions was prepared with the following chemical equation:



With the help of above chemical equation, the stoichiometric ratio of $\text{Ca}_{2(1-x)}\text{SiO}_4:\text{x}\text{Dy}^{3+}$ (where $\text{x} = 0.001, 0.005, 0.01, 0.05, 0.1, 0.2, 0.3, 0.4$ and $0.5 \text{ mol}\%$) have been synthesized by using solid state reaction method.

The stoichiometric quantities of CaO from egg shells, SiO_2 from rice husk and Dy_2O_3 (99.99%) from Sigma-Aldrich were weighed and grinded using an agate mortar. The grinded powders were pre-sintered at 1250°C for 8 h with several intermediate grindings and finally sintered at 1250°C for 8 h. The structure of the prepared powders were characterized by using X-ray diffractometer (RIGAKU, Miniflex-600) using $\text{Cu } K_\alpha$ radiation ($1.5406 \text{ \AA}$). The morphology of these phosphors was studied by scanning electron microscope (SEM) and elemental analysis was carried out using energy dispersive spectroscopy (EDS), by using CARL ZEISS EVO MA15 SEM. The excitation and emission spectra and decay curves were recorded using Edinburgh UV-vis-NIR (FLS 980) spectrometer equipped with xenon arc lamp (450 W) as an excitation source.

3. Results and discussion

3.1. Structure and morphology

The XRD patterns have been measured for undoped and Dy^{3+} -doped Ca_2SiO_4 phosphors. These XRD patterns along with their planes (hkl) have been shown in Fig. 1 for undoped and $0.5 \text{ mol}\%$ Dy^{3+} -doped

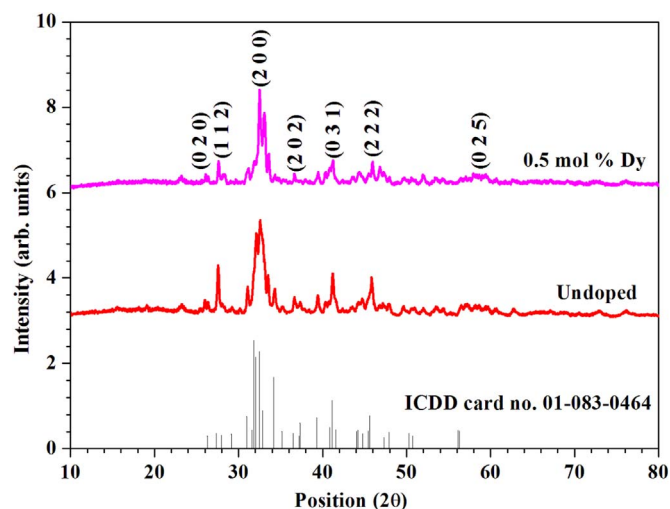


Fig. 1. X-Ray diffraction patterns and their (hkl) planes for undoped Ca_2SiO_4 and $0.5 \text{ mol}\%$ Dy^{3+} -doped Ca_2SiO_4 phosphor powders.

Ca_2SiO_4 phosphors. These patterns matches well with the standard ICDD card no. 01-083-0464. This clearly indicates that Dy^{3+} -doped Ca_2SiO_4 phosphors are crystallized in monoclinic structure with unit cell parameters of $a = 5.53 \text{ \AA}$; $b = 6.67 \text{ \AA}$; $c = 9.13 \text{ \AA}$; $\beta = 87.43^\circ$; and volume of $340 \text{ \AA}^3$. The particle size was calculated using the Scherrer's formula and is found to be 33 nm . The morphology of the 0.001 – $0.5 \text{ mol}\%$ Dy^{3+} -doped Ca_2SiO_4 phosphor has been characterized by SEM and is shown in Fig. 2 for $0.2 \text{ mol}\%$ Dy^{3+} -doped Ca_2SiO_4 phosphor. As can be seen from SEM images, particles of irregular in shape are distributed non-uniformly in synthesized phosphors. Further, elemental analysis has been carried out using EDS for all synthesized phosphors and is shown for $0.1 \text{ mol}\%$ Dy^{3+} -doped Ca_2SiO_4 phosphor (see Fig. 3). As can be seen from EDS spectrum, all the elements, i.e., calcium (Ca), dysprosium (Dy), silica (Si) and oxygen (O) have been identified in their respective proportions.

3.2. Luminescence spectra

The luminescence excitation spectra for the Dy^{3+} -doped Ca_2SiO_4 phosphors have been measured by monitoring the yellow emission at 573 nm . The excitation spectrum for $0.5 \text{ mol}\%$ Dy^{3+} -doped Ca_2SiO_4 phosphors are shown in Fig. 4 and exhibited three intense bands at 351 nm (${}^6\text{H}_{15/2} \rightarrow {}^6\text{P}_{7/2} + {}^4\text{M}_{15/2}$), 366 nm (${}^6\text{H}_{15/2} \rightarrow {}^6\text{P}_{5/2}$) and 387 nm

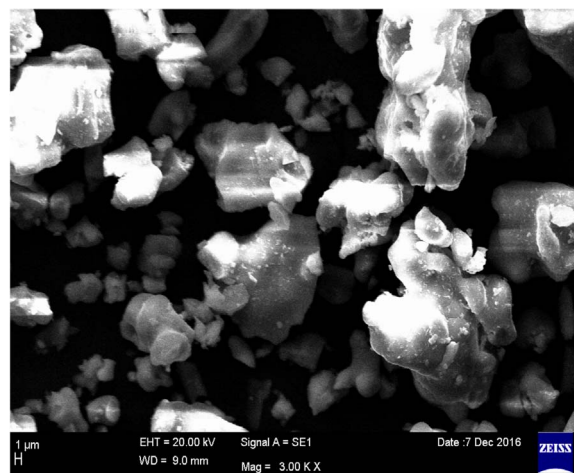
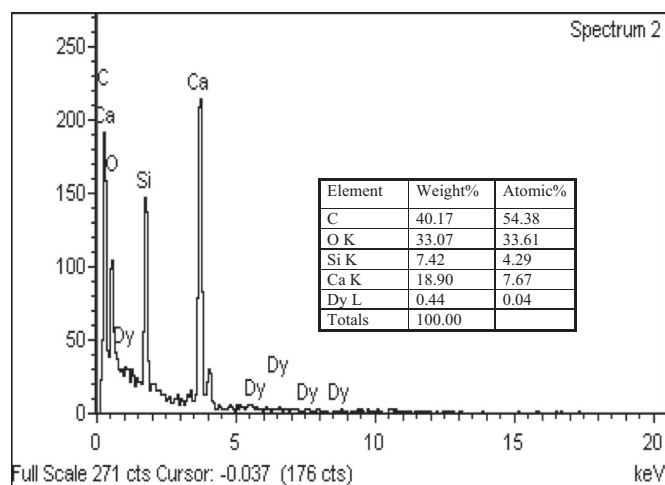
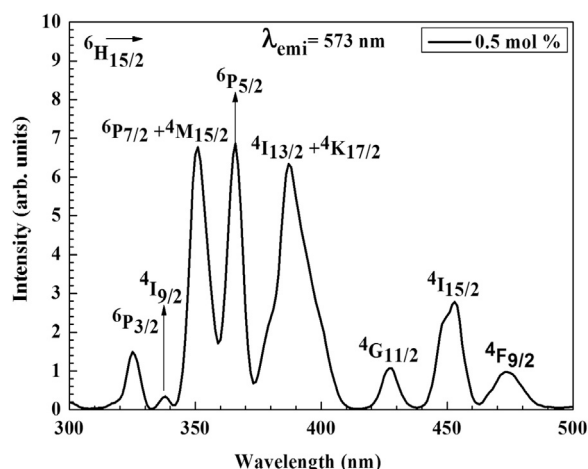
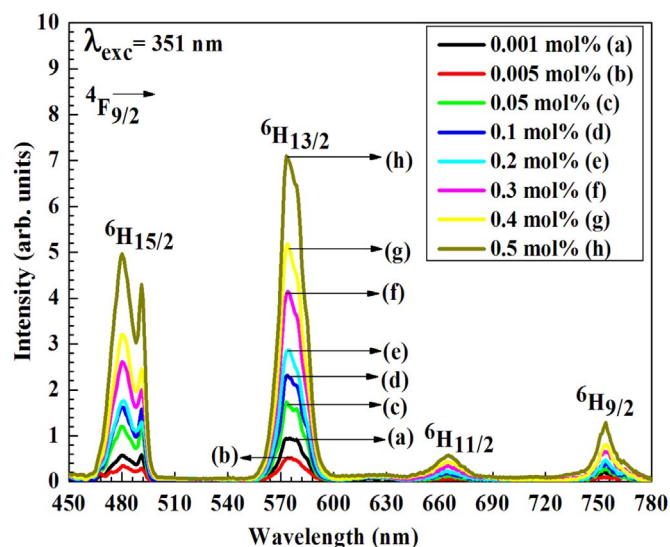


Fig. 2. Scanning electron micrograph of $0.2 \text{ mol}\%$ Dy^{3+} -doped Ca_2SiO_4 phosphors.

Fig. 3. Energy dispersive spectrum for 0.2 mol% Dy³⁺-doped Ca₂SiO₄ phosphors.Fig. 4. Excitation spectrum for 0.5 mol% Dy³⁺-doped Ca₂SiO₄ phosphors.

(⁶H_{15/2} → ⁴I_{13/2} + ⁴K_{17/2}) as well as two moderate bands at 325 nm (⁶H_{15/2} → ⁶P_{3/2}), 453 nm (⁶H_{15/2} → ⁴I_{15/2}) and three feeble bands at 337 nm (⁶H_{15/2} → ⁴I_{9/2}), 428 nm (⁶H_{15/2} → ⁴G_{11/2}) and 473 nm (⁶H_{15/2} → ⁴F_{9/2}). As can be seen from Fig. 4, high intensity bands are observed in the near UV and blue regions for Ca₂SiO₄:Dy³⁺ phosphor, which is prerequisite for the development of white-light emitting devices using commercial blue InGaN/GaN LED chip.

The emission spectra for Dy³⁺-doped Ca₂SiO₄ phosphors have been measured under 351 nm excitation. It consist of four emission peaks in blue (460–500 nm), yellow (555–600 nm), feeble red (650–685 nm) and deep red (740–780 nm) regions. These blue, yellow, feeble red and deep red emissions are assigned to the ⁴F_{9/2} → ⁶H_{15/2}, ⁴F_{9/2} → ⁶H_{13/2}, ⁴F_{9/2} → ⁶H_{11/2} and ⁴F_{9/2} → ⁶H_{9/2} transitions, respectively, as shown in Fig. 5. The blue emission corresponding to the ⁴F_{9/2} → ⁶H_{15/2} transition is magnetic-dipole in nature and its intensity is almost independent of local surroundings of the active ions. The yellow emission due to the ⁴F_{9/2} → ⁶H_{13/2} transition is hypersensitive and is allowed only at low symmetries with no inversion center. Therefore, in the present Ca₂SiO₄ phosphor, the Dy³⁺ ions are placed in such low symmetry sites of monoclinic and hence the ⁴F_{9/2} → ⁶H_{13/2} transition is prominent in their emission spectra. From the Fig. 5, it is found that the blue and yellow emission bands are splitted into two components due to the partial lifting of degeneracy by the crystal-field around the Dy³⁺ ion. With the increasing concentration, the crystal-field splitting increases due to the increase of surrounding Stark effect. As can be seen from the Fig. 5, it is observed that there is gradual increase in emission intensity with increase in Dy³⁺ ion concentration from 0.001 to 0.5 mol%

Fig. 5. Luminescence spectra for Dy³⁺-doped Ca₂SiO₄ phosphors.

[23,45,46]. It clearly indicates that the Ca²⁺ (r = 240 pm) ions are replaced by Dy³⁺ (r = 235 pm) ions which are the luminescence centers.

The (Y/B) intensity ratios of luminescence spectra have been evaluated. It is possible to generate white light by tuning of these colors (yellow and blue) emitted from the Dy³⁺ ions. In general, the Y/B ratio is sensitive to the chemical composition of the host but little changes are observed with the variation of Dy³⁺ ions concentration in the present study (See Table 1). From the Y/B ratios of Ca₂SiO₄:Dy³⁺ phosphors, the maximum value of 1.92 for 0.01 mol% and minimum value of 1.27 for 0.05 mol% of Dy³⁺ ions have been noticed. These small changes in Y/B ratios can be explained by the structural changes in the vicinity of Dy³⁺ ions [21,22,47,48]. The concentration dependent variations of Y/B ratios as well as excitation wavelength dependent variation of Y/B ratios are very small in the present Ca₂SiO₄:Dy³⁺ phosphors and hence the later experimental results are not shown. These results indicate that the Dy³⁺ ions occupy lower symmetry of monoclinic structure in the Ca₂SiO₄ powders as evidenced from XRD results.

The evaluated Y/B ratios of Ca₂SiO₄:Dy³⁺ phosphors are compared in Table 2 with those of other commercially available Dy³⁺-doped luminescent materials [13,14,27–30,36,48–51]. As the Y/B ratios of the present Ca₂SiO₄:Dy³⁺ phosphors are higher than those of Dy³⁺-doped Y₂CaZnO₅:Dy³⁺ (0.5 mol%) [14], Ba₅CaAl₄O₁₂:Dy³⁺ (0.10 mol%) [27], Sr₃Y(PO₄)₃:Dy³⁺ (4 mol%) [28], Ca₃Si₂O₇:Dy³⁺ (0.1 mol%) [29], Na₃SrB₅O₁₀:Dy³⁺ (7 mol%) [30], KY_(1-x)BO₃:Dy³⁺ (0.3 mol%) [36], NaLa(MoO₄)₂:Dy³⁺ (0.02 mol%) [49], YAl₃(BO₃)₄:Dy³⁺ (0.5 mol%) [50] and Ca₃Gd₇(PO₄)₄(SiO₄)₅O₂:Dy³⁺ (0.10 mol%) [51], whereas lower than those of BYF₅:Dy³⁺ (1 mol%) [13] and Lu₃Ga₅O₁₂:Dy³⁺ (2.0 mol%) [48].

Table 1

Y/B ratio, color coordinates (x, y) and lifetimes (τ, μs) for different concentrations (C, mol%) of Dy³⁺-doped Ca₂SiO₄ phosphors for 351 nm excitation.

S. no.	C	Y/B ratio	(x, y)	τ
1	0.001	1.52	(0.284, 0.286)	711
2	0.005	1.34	(0.239, 0.258)	739
3	0.01	1.92	(0.286, 0.288)	713
4	0.05	1.27	(0.264, 0.272)	712
5	0.1	1.31	(0.276, 0.292)	660
6	0.2	1.47	(0.299, 0.324)	573
7	0.3	1.47	(0.313, 0.338)	541
8	0.4	1.49	(0.317, 0.342)	515
9	0.5	1.33	(0.320, 0.349)	510

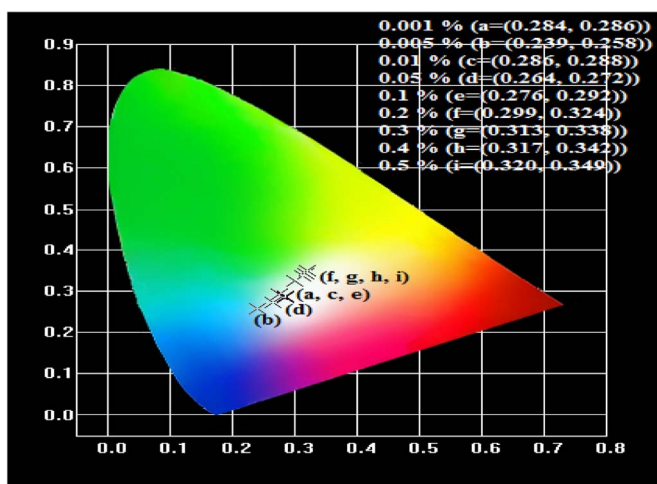
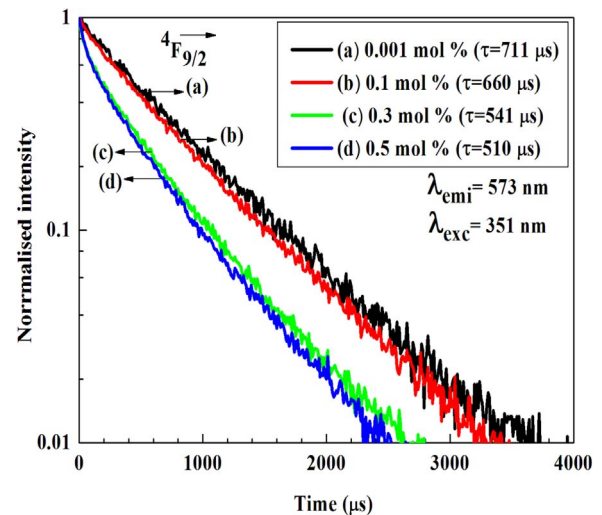
Table 2

Comparison of Y/B ratio, color coordinates (x, y) and lifetimes (τ , μ s) for Dy^{3+} : phosphors.

S. no.	Host	Y/B ratio	(x, y)	τ
1	$\text{Ca}_2\text{SiO}_4:0.5 \text{ Dy}^{3+}$ [Present work]	1.33	(0.320, 0.349)	510
2	$\text{BYF}_5:1 \text{ Dy}^{3+}$ [13]	1.98	(0.354, 0.402)	1300
3	$\text{Y}_2\text{CaZnO}_5:0.5 \text{ Dy}^{3+}$ [14]	1.12	(0.378, 0.438)	458
4	$\text{YVO}_4:1 \text{ Dy}^{3+}$ [24]	–	(0.39, 0.44)	290
5	$\text{Ca}_9\text{La}(\text{PO}_4)_5(\text{SiO}_4)\text{Cl}_2:0.20 \text{ Dy}^{3+}$ [25]	–	–	477
6	$\text{Ba}_{10}\text{F}_2(\text{PO}_4)_6:0.40 \text{ Dy}^{3+}$ [26]	–	(0.300, 0.357)	688
7	$\text{Ba}_5\text{CaAl}_4\text{O}_{12}:0.10 \text{ Dy}^{3+}$ [27]	1.09	(0.333, 0.361)	–
8	$\text{Sr}_3\text{Y}(\text{PO}_4)_3:4 \text{ Dy}^{3+}$ [28]	0.876	(0.330, 0.366)	–
9	$\text{Ca}_3\text{Si}_2\text{O}_7:0.1 \text{ Dy}^{3+}$ [29]	0.765	(0.33, 0.33)	–
10	$\text{Na}_3\text{SrB}_5\text{O}_{10}:7 \text{ Dy}^{3+}$ [30]	0.89	(0.305, 0.317)	570
11	$\text{NaPbB}_5\text{O}_9:0.02 \text{ Dy}^{3+}$ [31]	–	(0.33, 0.36)	349
12	$\text{Li}_2\text{SrSiO}_4:0.5 \text{ Dy}^{3+}$ [32]	–	(0.259, 0.289)	902
13	$\text{Sr}_3\text{Bi}(\text{PO}_4)_3:0.07 \text{ Dy}^{3+}$ [33]	–	(0.327, 0.353)	–
14	$\text{KY}_{(1-x)}\text{BO}_3:0.3 \text{ Dy}^{3+}$ [36]	1.20	(0.29, 0.32)	130
15	$\text{Lu}_3\text{Ga}_5\text{O}_{12}:2.0 \text{ Dy}^{3+}$ [48]	2.57	(0.386, 0.391)	1450
16	$\text{NaLa}(\text{MoO}_4)_2:0.02 \text{ Dy}^{3+}$ [49]	1.63	(0.336, 0.392)	–
17	$\text{YAl}_3(\text{BO}_3)_4:0.5 \text{ Dy}^{3+}$ [50]	0.9462	(0.2838, 0.3101)	565
18	$\text{Ca}_3\text{Gd}_7(\text{PO}_4)(\text{SiO}_4)_5\text{O}_2:0.10 \text{ Dy}^{3+}$ [51]	1.14	(0.328, 0.343)	–
19	$\text{Sr}_2\text{SiO}_4:6 \text{ Dy}^{3+}$ [53]	–	(0.351, 0.389)	59.3
20	$\text{KLa}(\text{PO}_4)_4:0.5 \text{ Dy}^{3+}$ [54]	–	(0.215, 0.221)	870
21	$\text{Gd}(\text{BO}_2)_3:0.06 \text{ Dy}^{3+}$ [55]	–	–	12
22	$\text{Ca}_9\text{Y}(\text{VO}_4)_3:0.1 \text{ Dy}^{3+}$ [56]	–	(0.339, 0.417)	226
23	$\text{Sr}_2\text{MgSi}_2\text{O}_7:0.5 \text{ Dy}^{3+}$ [58]	–	(0.310, 0.328)	–
24	$\text{ZnWO}_4:0.5 \text{ Dy}^{3+}$ [59]	–	(0.2681, 0.3091)	–
25	$\text{Sr}_3\text{Gd}(\text{PO}_4)_3:0.08 \text{ Dy}^{3+}$ [60]	–	(0.2461, 0.2784)	–
26	$\text{LaBSiO}_5:0.008 \text{ Dy}^{3+}$ [61]	–	(0.3145, 0.3943)	–

3.3. Color coordinates

The color coordinates are necessary to know the nature of color emitted by phosphors. The color coordinates have been determined using the emission spectra and are in turn used to calculate the correlated color temperatures (CCTs) [52]. The evaluated color coordinates for the $\text{Ca}_2\text{SiO}_4:\text{Dy}^{3+}$ phosphors are shown in Fig. 6 and also presented in Table 1. From Fig. 6, it is observed that the color coordinates of present phosphors fall within the white light region and are close to the “ideal white light” ($x = 0.33$, $y = 0.33$) in the chromaticity diagram. The color coordinates for the present and reported systems are compared in Table 2. As can be seen, the color coordinates obtained for $\text{Ca}_2\text{SiO}_4:\text{Dy}^{3+}$ are comparable to those of the reported systems [13,14,24,26–33,36,48–51,53,54,56,58–61]. The CCT values for $\text{Ca}_2\text{SiO}_4:\text{Dy}^{3+}$ phosphors are found to vary from 4989 to 6369 K. From these observations, it is worth noting that the present phosphors made

Fig. 6. CIE chromaticity graph for Dy^{3+} -doped Ca_2SiO_4 phosphors.Fig. 7. Decay curves for $^4\text{F}_{9/2}$ level of Dy^{3+} ions in Ca_2SiO_4 phosphors.

from the agriculture waste materials (rice husk and egg shells), exhibit white light luminescence [24,31,53] and are comparable to those of other reported phosphors [31,49,55]. Hence, the $\text{Ca}_2\text{SiO}_4:\text{Dy}^{3+}$ phosphors synthesized from the utilization of agriculture waste could be useful for the fabrication of low cost white light-emitting devices.

3.4. Luminescent decay curves

The concentration dependent luminescent decay curves for the $^4\text{F}_{9/2}$ luminescent level of Dy^{3+} ions in Ca_2SiO_4 phosphors measured under 351 nm excitation by monitoring 573 nm emission, are presented in Fig. 7. For lower concentrations (0.001, 0.005, 0.01, 0.05, 0.1 and 0.2 mol%), the decay curves are found to be nearly single exponential and as the concentration increases they turned to non-exponential nature (≥ 0.3 mol%). Hence, the effective lifetimes (τ_{eff}) for both the near single exponential and non-exponential decay curves have been evaluated using the expression,

$$\tau_{\text{eff}} = \frac{\int t I(t) dt}{\int I(t) dt} \quad (1)$$

The effective lifetimes for $^4\text{F}_{9/2}$ level of Dy^{3+} ions are found to be 711, 739, 713, 712, 660, 573, 541, 515 and 510 μ s for 0.001, 0.005, 0.01, 0.05, 0.1, 0.2, 0.3, 0.4 and 0.5 mol% Dy^{3+} -doped Ca_2SiO_4 phosphors, respectively, and are shown in Table 1. It is found that the lifetime for the $^4\text{F}_{9/2}$ level increases upto 0.005 mol% Dy^{3+} concentration and then decreases. The increase in lifetime with increase in Dy^{3+} concentration from 0.001 to 0.005 mol% is due to the process of reabsorption. It is well known that for lower concentrations, the lifetime is higher and increases slightly with increase in concentration due to reabsorption of emitted photons. Beyond certain value, the lifetime starts to decrease with increase in concentration due to energy transfer between the luminescent ions. As can be seen from Table 1, in the present work also, lifetime increases with increase in concentration and then starts to decrease. The effective lifetimes are found to decrease with increase in Dy^{3+} concentration due to energy transfer between Dy^{3+} ions through cross-relaxation processes [13,48] and/or energy transfer to luminescence quenching centers (impurities). Similar decreasing trend with increase in Dy^{3+} concentration was reported for other Dy^{3+} : phosphors [25,26,30,31,55].

Lifetimes for the $^4\text{F}_{9/2}$ level of Dy^{3+} ions in different phosphors have been compared in Table 2. The effective lifetimes observed for the present $\text{Ca}_2\text{SiO}_4:\text{Dy}^{3+}$ phosphors are found to be higher than those of $\text{BYF}_5:\text{Dy}^{3+}$ (1 mol%) [13], $\text{Ba}_{10}\text{F}_2(\text{PO}_4)_6:\text{Dy}^{3+}$ (0.40 mol%) [26], $\text{Na}_3\text{SrB}_5\text{O}_{10}:\text{Dy}^{3+}$ (7 mol%) [30], $\text{Li}_2\text{SrSiO}_4:\text{Dy}^{3+}$ (0.5 mol%) [32],

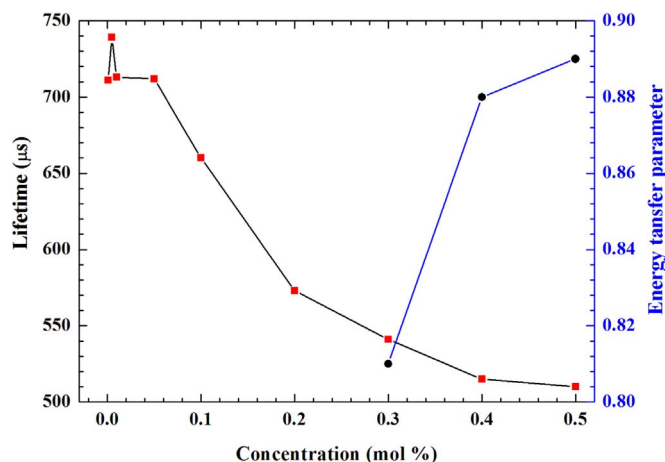


Fig. 8. The concentration dependent lifetime and energy transfer parameter for Dy^{3+} -doped Ca_2SiO_4 phosphors.

$\text{Lu}_3\text{Ga}_5\text{O}_{12}:\text{Dy}^{3+}$ (2.0 mol%) [48], $\text{YAl}_3(\text{BO}_3)_4:\text{Dy}^{3+}$ (0.5 mol%) [50], $\text{KLa}(\text{PO}_3)_4:\text{Dy}^{3+}$ (0.5 mol%) [54], whereas lower than those of $\text{Y}_2\text{CaZnO}_5:\text{Dy}^{3+}$ (0.5 mol%) [14], $\text{YVO}_4:\text{Dy}^{3+}$ (1 mol%) [24], $\text{Ca}_9\text{La}(\text{PO}_4)_5(\text{SiO}_4)\text{Cl}_2:\text{Dy}^{3+}$ (0.20 mol%) [25], $\text{NaPbB}_5\text{O}_9:\text{Dy}^{3+}$ (0.02 mol%) [31], $\text{KY}_{(1-x)}\text{BO}_3:\text{Dy}^{3+}$ (0.3 mol%) [36], $\text{Sr}_2\text{SiO}_4:\text{Dy}^{3+}$ (6 mol%) [53], $\text{Gd}(\text{BO}_2)_3:\text{Dy}^{3+}$ (0.06 mol%) [55], $\text{Ca}_9\text{Y}(\text{VO}_4)_3:\text{Dy}^{3+}$ (0.1 mol%) [56].

The non-exponential decay curves of Dy^{3+} -doped Ca_2SiO_4 powders arises due to multipolar (dipole-dipole, dipole-quadrupole, quadrupole-quadrupole) interactions between active ions. To know the type of the multipolar interaction in our phosphors, Inokuti-Hirayama (IH) [57] model has been adopted. According to this model, the decay of luminescence intensity, $I(t)$, is

$$I(t) = I_0 \exp \left\{ -\frac{t}{\tau_0} - Q \left(\frac{t}{\tau_0} \right)^{3/5} \right\} \quad (2)$$

where 't' is the time after excitation, τ_0 is the intrinsic decay time of the donors in the absence of acceptors. Q is the energy transfer parameter, which depends on dipole-dipole ($S = 6$), dipole-quadrupole ($S = 8$) and quadrupole-quadrupole ($S = 10$) interactions. As can be seen from Fig. 7, the best fit has been obtained for $S = 6$, which indicates that the mechanism of energy transfer between Dy^{3+} ions is of dipole-dipole type. The energy transfer parameter (Q) increases with increase in concentration of Dy^{3+} ions. The lifetime values decreases with increase in Dy^{3+} ions concentrations (see Fig. 8).

4. Conclusions

A series of Dy^{3+} -doped Ca_2SiO_4 phosphors have been successfully synthesized from agricultural waste by the solid-state reaction method and investigated their structural, morphological, and luminescence properties. The synthesized Dy^{3+} -doped Ca_2SiO_4 phosphors are crystallized in monoclinic structure. The morphology of the Dy^{3+} -doped Ca_2SiO_4 phosphors are of irregular in shape with different sizes. The Y/B ratios are found to vary from 1.52 to 1.33 when Dy^{3+} concentrations vary from 0.001 mol% to 0.5 mol%. The color co-ordinates obtained from emission spectra are found to fall within the white light zone. The fluorescence decay curves are found to be nearly single exponential in nature for lower Dy^{3+} ions concentration and turns into non-exponential for higher concentrations (≥ 0.3 mol%) due to energy transfer between Dy^{3+} ions through dipole-dipole interactions. The results obtained indicate that the $\text{Ca}_2\text{SiO}_4:\text{Dy}^{3+}$ phosphors are suitable for fabrication of low cost white light emitting devices.

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